Хуваагдах чанар ба Модулын арифметик
Семинар-5

Exercises

1. Does 17 divide each of these numbers?
a) 68 **b)** 84 **c)** 357 **d)** 1001

2. Prove that if a is an integer other than 0, then
a) 1 divides a. **b)** a divides 0.

3. Prove that part (*ii*) of Theorem 1 is true.

4. Prove that part (*iii*) of Theorem 1 is true.

5. Show that if $a \mid b$ and $b \mid a$, where a and b are integers, then $a = b$ or $a = -b$.

6. Show that if a, b, c, and d are integers, where $a \neq 0$, such that $a \mid c$ and $b \mid d$, then $ab \mid cd$.

7. Show that if a, b, and c are integers, where $a \neq 0$ and $c \neq 0$, such that $ac \mid bc$, then $a \mid b$.

8. Prove or disprove that if $a \mid bc$, where a, b, and c are positive integers and $a \neq 0$, then $a \mid b$ or $a \mid c$.

9. Prove that if a and b are integers and a divides b, then a is odd or b is even.

10. Prove that if a and b are nonzero integers, a divides b, and $a + b$ is odd, then a is odd.

11. Prove that if a is an integer that is not divisible by 3, then $(a + 1)(a + 2)$ is divisible by 3.

12. Prove that if a is a positive integer, then 4 does not divide $a^2 + 2$.

13. What are the quotient and remainder when
a) 19 is divided by 7? **b)** -111 is divided by 11?
c) 789 is divided by 23? **d)** 1001 is divided by 13?
e) 0 is divided by 19? **f)** 3 is divided by 5?
g) -1 is divided by 3? **h)** 4 is divided by 1?

14. What are the quotient and remainder when
a) 44 is divided by 8?
b) 777 is divided by 21?
c) -123 is divided by 19?
d) -1 is divided by 23?
e) -2002 is divided by 87?
f) 0 is divided by 17?
g) 1,234,567 is divided by 1001?
h) -100 is divided by 101?

15. What time does a 12-hour clock read
a) 80 hours after it reads 11:00?
b) 40 hours before it reads 12:00?
c) 100 hours after it reads 6:00?

16. What time does a 24-hour clock read
a) 100 hours after it reads 2:00?
b) 45 hours before it reads 12:00?
c) 168 hours after it reads 19:00?

17. Suppose that a and b are integers, $a \equiv 4 \pmod{13}$, and $b \equiv 9 \pmod{13}$. Find the integer c with $0 \leq c \leq 12$ such that
a) $c \equiv 9a \pmod{13}$.
b) $c \equiv 11b \pmod{13}$.
c) $c \equiv a + b \pmod{13}$.
d) $c \equiv 2a + 3b \pmod{13}$.
e) $c \equiv a^2 + b^2 \pmod{13}$.
f) $c \equiv a^3 - b^3 \pmod{13}$.

18. Suppose that a and b are integers, $a \equiv 11 \pmod{19}$, and $b \equiv 3 \pmod{19}$. Find the integer c with $0 \leq c \leq 18$ such that
a) $c \equiv 13a \pmod{19}$.
b) $c \equiv 8b \pmod{19}$.
c) $c \equiv a - b \pmod{19}$.
d) $c \equiv 7a + 3b \pmod{19}$.
e) $c \equiv 2a^2 + 3b^2 \pmod{19}$.
f) $c \equiv a^3 + 4b^3 \pmod{19}$.

19. Show that if a and d are positive integers, then $(-a)$ **div** $d = -a$ **div** d if and only if d divides a.

20. Prove or disprove that if a, b, and d are integers with $d > 0$, then $(a + b)$ **div** $d = a$ **div** $d + b$ **div** d.

21. Let m be a positive integer. Show that $a \equiv b \pmod{m}$ if a **mod** $m = b$ **mod** m.

22. Let m be a positive integer. Show that a **mod** $m = b$ **mod** m if $a \equiv b \pmod{m}$.

23. Show that if n and k are positive integers, then $\lceil n/k \rceil = \lfloor (n-1)/k \rfloor + 1$.

24. Show that if a is an integer and d is an integer greater than 1, then the quotient and remainder obtained when a is divided by d are $\lfloor a/d \rfloor$ and $a - d\lfloor a/d \rfloor$, respectively.

25. Find a formula for the integer with smallest absolute value that is congruent to an integer a modulo m, where m is a positive integer.

26. Evaluate these quantities.

a) -17 **mod** 2 **b)** 144 **mod** 7
c) -101 **mod** 13 **d)** 199 **mod** 19

27. Evaluate these quantities.

a) 13 **mod** 3 **b)** -97 **mod** 11
c) 155 **mod** 19 **d)** -221 **mod** 23

28. Find a **div** m and a **mod** m when

a) $a = -111$, $m = 99$.
b) $a = -9999$, $m = 101$.
c) $a = 10299$, $m = 999$.
d) $a = 123456$, $m = 1001$.

29. Find a **div** m and a **mod** m when

a) $a = 228$, $m = 119$.
b) $a = 9009$, $m = 223$.
c) $a = -10101$, $m = 333$.
d) $a = -765432$, $m = 38271$.

30. Find the integer a such that

a) $a \equiv 43 \pmod{23}$ and $-22 \leq a \leq 0$.
b) $a \equiv 17 \pmod{29}$ and $-14 \leq a \leq 14$.
c) $a \equiv -11 \pmod{21}$ and $90 \leq a \leq 110$.

31. Find the integer a such that

a) $a \equiv -15 \pmod{27}$ and $-26 \leq a \leq 0$.
b) $a \equiv 24 \pmod{31}$ and $-15 \leq a \leq 15$.
c) $a \equiv 99 \pmod{41}$ and $100 \leq a \leq 140$.

32. List five integers that are congruent to 4 modulo 12.

33. List all integers between -100 and 100 that are congruent to -1 modulo 25.

34. Decide whether each of these integers is congruent to 3 modulo 7.

a) 37 **b)** 66
c) -17 **d)** -67

35. Decide whether each of these integers is congruent to 5 modulo 17.

a) 80 **b)** 103
c) -29 **d)** -122

36. Find each of these values.

a) $(177$ **mod** $31 + 270$ **mod** $31)$ **mod** 31
b) $(177$ **mod** $31 \cdot 270$ **mod** $31)$ **mod** 31

37. Find each of these values.

a) $(-133$ **mod** $23 + 261$ **mod** $23)$ **mod** 23
b) $(457$ **mod** $23 \cdot 182$ **mod** $23)$ **mod** 23

38. Find each of these values.

a) $(19^2$ **mod** $41)$ **mod** 9
b) $(32^3$ **mod** $13)^2$ **mod** 11
c) $(7^3$ **mod** $23)^2$ **mod** 31
d) $(21^2$ **mod** $15)^3$ **mod** 22

39. Find each of these values.

a) $(99^2$ **mod** $32)^3$ **mod** 15
b) $(3^4$ **mod** $17)^2$ **mod** 11
c) $(19^3$ **mod** $23)^2$ **mod** 31
d) $(89^3$ **mod** $79)^4$ **mod** 26

40. Show that if $a \equiv b \pmod{m}$ and $c \equiv d \pmod{m}$, where a, b, c, d, and m are integers with $m \geq 2$, then $a - c \equiv b - d \pmod{m}$.

41. Show that if $n \mid m$, where n and m are integers greater than 1, and if $a \equiv b \pmod{m}$, where a and b are integers, then $a \equiv b \pmod{n}$.

☞ **42.** Show that if a, b, c, and m are integers such that $m \geq 2$, $c > 0$, and $a \equiv b \pmod{m}$, then $ac \equiv bc \pmod{mc}$.

43. Find counterexamples to each of these statements about congruences.

a) If $ac \equiv bc \pmod{m}$, where a, b, c, and m are integers with $m \geq 2$, then $a \equiv b \pmod{m}$.
b) If $a \equiv b \pmod{m}$ and $c \equiv d \pmod{m}$, where a, b, c, d, and m are integers with c and d positive and $m \geq 2$, then $a^c \equiv b^d \pmod{m}$.

44. Show that if n is an integer then $n^2 \equiv 0$ or $1 \pmod{4}$.

45. Use Exercise 44 to show that if m is a positive integer of the form $4k + 3$ for some nonnegative integer k, then m is not the sum of the squares of two integers.

46. Prove that if n is an odd positive integer, then $n^2 \equiv 1 \pmod{8}$.

47. Show that if a, b, k, and m are integers such that $k \geq 1$, $m \geq 2$, and $a \equiv b \pmod{m}$, then $a^k \equiv b^k \pmod{m}$.

48. Show that $\mathbf{Z}_m$ with addition modulo m, where $m \geq 2$ is an integer, satisfies the closure, associative, and commutative properties, 0 is an additive identity, and for every nonzero $a \in \mathbf{Z}_m$, $m - a$ is an inverse of a modulo m.

49. Show that $\mathbf{Z}_m$ with multiplication modulo m, where $m \geq 2$ is an integer, satisfies the closure, associative, and commutativity properties, and 1 is a multiplicative identity.

50. Show that the distributive property of multiplication over addition holds for $\mathbf{Z}_m$, where $m \geq 2$ is an integer.

51. Write out the addition and multiplication tables for $\mathbf{Z}_5$ (where by addition and multiplication we mean $+_5$ and $\cdot_5$).

52. Write out the addition and multiplication tables for $\mathbf{Z}_6$ (where by addition and multiplication we mean $+_6$ and $\cdot_6$).

53. Determine whether each of the functions $f(a) = a$ **div** d and $g(a) = a$ **mod** d, where d is a fixed positive integer, from the set of integers to the set of integers, is one-to-one, and determine whether each of these functions is onto.

Exercises

1. Convert the decimal expansion of each of these integers to a binary expansion.
a) 231 **b)** 4532 **c)** 97644

2. Convert the decimal expansion of each of these integers to a binary expansion.
a) 321 **b)** 1023 **c)** 100632

3. Convert the binary expansion of each of these integers to a decimal expansion.
a) $(1\ 1111)_2$ **b)** $(10\ 0000\ 0001)_2$
c) $(1\ 0101\ 0101)_2$ **d)** $(110\ 1001\ 0001\ 0000)_2$

4. Convert the binary expansion of each of these integers to a decimal expansion.
a) $(1\ 1011)_2$ **b)** $(10\ 1011\ 0101)_2$
c) $(11\ 1011\ 1110)_2$ **d)** $(111\ 1100\ 0001\ 1111)_2$

5. Convert the octal expansion of each of these integers to a binary expansion.
a) $(572)_8$ **b)** $(1604)_8$
c) $(423)_8$ **d)** $(2417)_8$

6. Convert the binary expansion of each of these integers to an octal expansion.
a) $(1111\ 0111)_2$
b) $(1010\ 1010\ 1010)_2$
c) $(111\ 0111\ 0111\ 0111)_2$
d) $(101\ 0101\ 0101\ 0101)_2$

7. Convert the hexadecimal expansion of each of these integers to a binary expansion.
a) $(80E)_{16}$ **b)** $(135AB)_{16}$
c) $(ABBA)_{16}$ **d)** $(DEFACED)_{16}$

8. Convert $(BADFACED)_{16}$ from its hexadecimal expansion to its binary expansion.

9. Convert $(ABCDEF)_{16}$ from its hexadecimal expansion to its binary expansion.

10. Convert each of the integers in Exercise 6 from a binary expansion to a hexadecimal expansion.

11. Convert $(1011\ 0111\ 1011)_2$ from its binary expansion to its hexadecimal expansion.

12. Convert $(1\ 1000\ 0110\ 0011)_2$ from its binary expansion to its hexadecimal expansion.

13. Show that the hexadecimal expansion of a positive integer can be obtained from its binary expansion by grouping together blocks of four binary digits, adding initial zeros if necessary, and translating each block of four binary digits into a single hexadecimal digit.

14. Show that the binary expansion of a positive integer can be obtained from its hexadecimal expansion by translating each hexadecimal digit into a block of four binary digits.

15. Show that the octal expansion of a positive integer can be obtained from its binary expansion by grouping together blocks of three binary digits, adding initial zeros if necessary, and translating each block of three binary digits into a single octal digit.

16. Show that the binary expansion of a positive integer can be obtained from its octal expansion by translating each octal digit into a block of three binary digits.

17. Convert $(7345321)_8$ to its binary expansion and $(10\ 1011\ 1011)_2$ to its octal expansion.

18. Give a procedure for converting from the hexadecimal expansion of an integer to its octal expansion using binary notation as an intermediate step.

19. Give a procedure for converting from the octal expansion of an integer to its hexadecimal expansion using binary notation as an intermediate step.

20. Explain how to convert from binary to base 64 expansions and from base 64 expansions to binary expansions and from octal to base 64 expansions and from base 64 expansions to octal expansions.

21. Find the sum and the product of each of these pairs of numbers. Express your answers as a binary expansion.
a) $(100\ 0111)_2$, $(111\ 0111)_2$
b) $(1110\ 1111)_2$, $(1011\ 1101)_2$
c) $(10\ 1010\ 1010)_2$, $(1\ 1111\ 0000)_2$
d) $(10\ 0000\ 0001)_2$, $(11\ 1111\ 1111)_2$

22. Find the sum and product of each of these pairs of numbers. Express your answers as a base 3 expansion.
a) $(112)_3$, $(210)_3$
b) $(2112)_3$, $(12021)_3$
c) $(20001)_3$, $(1111)_3$
d) $(120021)_3$, $(2002)_3$

23. Find the sum and product of each of these pairs of numbers. Express your answers as an octal expansion.
a) $(763)_8$, $(147)_8$
b) $(6001)_8$, $(272)_8$
c) $(1111)_8$, $(777)_8$
d) $(54321)_8$, $(3456)_8$

24. Find the sum and product of each of these pairs of numbers. Express your answers as a hexadecimal expansion.
a) $(1AE)_{16}$, $(BBC)_{16}$
b) $(20CBA)_{16}$, $(A01)_{16}$
c) $(ABCDE)_{16}$, $(1111)_{16}$
d) $(E0000E)_{16}$, $(BAAA)_{16}$

25. Use Algorithm 5 to find 7^{644} **mod** 645.

26. Use Algorithm 5 to find 11^{644} **mod** 645.

27. Use Algorithm 5 to find 3^{2003} **mod** 99.

28. Use Algorithm 5 to find 123^{1001} **mod** 101.

29. Show that every positive integer can be represented uniquely as the sum of distinct powers of 2. [*Hint:* Consider binary expansions of integers.]

30. It can be shown that every integer can be uniquely represented in the form

$$e_k 3^k + e_{k-1} 3^{k-1} + \cdots + e_1 3 + e_0,$$

where $e_j = -1, 0,$ or 1 for $j = 0, 1, 2, \ldots, k$. Expansions of this type are called **balanced ternary expansions**. Find the balanced ternary expansions of
a) 5. **b)** 13. **c)** 37. **d)** 79.

31. Show that a positive integer is divisible by 3 if and only if the sum of its decimal digits is divisible by 3.

32. Show that a positive integer is divisible by 11 if and only if the difference of the sum of its decimal digits in even-numbered positions and the sum of its decimal digits in odd-numbered positions is divisible by 11.

33. Show that a positive integer is divisible by 3 if and only if the difference of the sum of its binary digits in even-numbered positions and the sum of its binary digits in odd-numbered positions is divisible by 3.

34. Determine how we can use the decimal expansion of an integer n to determine whether n is divisible by

a) 2 **b)** 5 **c)** 10

35. Determine how we can use the decimal expansion of an integer n to determine whether n is divisible by

a) 4 **b)** 25 **c)** 20

36. Suppose that n and b are positive integers with $b \geq 2$ and the base b expansion of n is $n = (a_m a_{m-1} \dots a_1 a_0)_b$. Find the base b expansion of

a) bn. **b)** $b^2 n$;

c) $\lfloor n/b \rfloor$, **d)** $\lfloor n/b^2 \rfloor$.

37. Prove that if n and b are positive integers with $b \geq 2$ the base b representation of n has $\lfloor \log_b n \rfloor + 1$ digits.

38. Find the decimal expansion of the number with the n-digit base seven expansion $(111 \dots 111)_7$ (with n 1's). [*Hint*: Use the formula for the sum of the terms of a geometric progression.]

39. Find the decimal expansion of the number with the $3n$ bit binary expansion $(101101....101101)_2$ (so that the binary expansion is made of n copies of 101). [*Hint*: Use the formula for the sum of the terms of a geometric progression.]

One's complement representations of integers are used to simplify computer arithmetic. To represent positive and negative integers with absolute value less than 2^{n-1}, a total of n bits is used. The leftmost bit is used to represent the sign. A 0 bit in this position is used for positive integers, and a 1 bit in this position is used for negative integers. For positive integers, the remaining bits are identical to the binary expansion of the integer. For negative integers, the remaining bits are obtained by first finding the binary expansion of the absolute value of the integer, and then taking the complement of each of these bits, where the complement of a 1 is a 0 and the complement of a 0 is a 1.

40. Find the one's complement representations, using bit strings of length six, of the following integers.

a) 22 **b)** 31 **c)** -7 **d)** -19

41. What integer does each of the following one's complement representations of length five represent?

a) 11001 **b)** 01101

c) 10001 **d)** 11111

42. If m is a positive integer less than 2^{n-1}, how is the one's complement representation of $-m$ obtained from the one's complement of m, when bit strings of length n are used?

43. How is the one's complement representation of the sum of two integers obtained from the one's complement representations of these integers?

44. How is the one's complement representation of the difference of two integers obtained from the one's complement representations of these integers?

45. Show that the integer m with one's complement representation $(a_{n-1}a_{n-2} \dots a_1 a_0)$ can be found using the equation $m = -a_{n-1}(2^{n-1} - 1) + a_{n-2}2^{n-2} + \cdots + a_1 \cdot 2 + a_0$.

Two's complement representations of integers are also used to simplify computer arithmetic and are used more commonly than one's complement representations. To represent an integer x with $-2^{n-1} \leq x \leq 2^{n-1} - 1$ for a specified positive integer n, a total of n bits is used. The leftmost bit is used to represent the sign. A 0 bit in this position is used for positive integers, and a 1 bit in this position is used for negative integers, just as in one's complement expansions. For a positive integer, the remaining bits are identical to the binary expansion of the integer. For a negative integer, the remaining bits are the bits of the binary expansion of $2^{n-1} - |x|$. Two's complement expansions of integers are often used by computers because addition and subtraction of integers can be performed easily using these expansions, where these integers can be either positive or negative.

46. Answer Exercise 40, but this time find the two's complement expansion using bit strings of length six.

47. Answer Exercise 41 if each expansion is a two's complement expansion of length five.

48. Answer Exercise 42 for two's complement expansions.

49. Answer Exercise 43 for two's complement expansions.

50. Answer Exercise 44 for two's complement expansions.

51. Show that the integer m with two's complement representation $(a_{n-1}a_{n-2} \dots a_1 a_0)$ can be found using the equation $m = -a_{n-1} \cdot 2^{n-1} + a_{n-2}2^{n-2} + \cdots + a_1 \cdot 2 + a_0$.

52. Give a simple algorithm for forming the two's complement representation of an integer from its one's complement representation.

53. Sometimes integers are encoded by using four-digit binary expansions to represent each decimal digit. This produces the **binary coded decimal** form of the integer. For instance, 791 is encoded in this way by 011110010001. How many bits are required to represent a number with n decimal digits using this type of encoding?

A **Cantor expansion** is a sum of the form

$$a_n n! + a_{n-1}(n-1)! + \cdots + a_2 2! + a_1 1!,$$

where a_i is an integer with $0 \leq a_i \leq i$ for $i = 1, 2, \dots, n$.

54. Find the Cantor expansions of

a) 2. **b)** 7.

c) 19. **d)** 87.

e) 1000. **f)** 1,000,000.

***55.** Describe an algorithm that finds the Cantor expansion of an integer.

***56.** Describe an algorithm to add two integers from their Cantor expansions.

57. Add $(10111)_2$ and $(11010)_2$ by working through each step of the algorithm for addition given in the text.

58. Multiply $(1110)_2$ and $(1010)_2$ by working through each step of the algorithm for multiplication given in the text.

59. Describe an algorithm for finding the difference of two binary expansions.

60. Estimate the number of bit operations used to subtract two binary expansions.

61. Devise an algorithm that, given the binary expansions of the integers a and b, determines whether $a > b$, $a = b$, or $a < b$.

62. How many bit operations does the comparison algorithm from Exercise 61 use when the larger of a and b has n bits in its binary expansion?

63. Estimate the complexity of Algorithm 1 for finding the base b expansion of an integer n in terms of the number of divisions used.

***64.** Show that Algorithm 5 uses $O((\log m)^2 \log n)$ bit operations to find $b^n \bmod m$.

65. Show that Algorithm 4 uses $O(q \log a)$ bit operations, assuming that $a > d$.